Principal-Series Blocks on the Celestial Sphere: Conical Reduction, Shadow Symmetry, and Exact Moments

Daniel Toupin
Golden Physics Project
`goldenphysics.org`

August 2026 — v2, superseding `kinematic_block_v1.1`

Abstract

This paper collects the special-function structure of scalar principal-series blocks on the celestial sphere, and re-anchors it as a toolkit for the partial-wave decomposition of unitarity cuts. Six results are stated and proved, each verified to at least 25 digits. The chiral block $k_h(z) = z^h\, {}_2F_1(h, h; 2h; z)$ reduces exactly to a Legendre function of the second kind, $k_h(z) = c(h)\, Q_{h-1}(2/z - 1)$, in the off-cut branch appropriate to argument greater than one; the branch choice is not cosmetic, and the identity fails by roughly 50% in the on-cut convention. The shadow involution $\Delta \mapsto 2 - \Delta$ becomes the classical Legendre degree symmetry $\nu \mapsto -\nu - 1$, whose connection formula $Q_{-\nu-1} - Q_\nu = i\pi \tanh(\pi\tau) P_\nu$ exhibits the shadow-odd part of the block as a Mehler–Fock kernel. The reduction prefactor has the exact modulus $|c(\lambda)|^2 = (2\lambda/\pi)\coth(\pi\lambda/2)$. On physical kinematics the full block is bounded in λ, with envelope $|1 - z|^{-1/2}$, since the chiral exponential rates cancel by conjugate oddness. We give a short closed-form proof, previously only a numerical observation, of the first moment

$$\frac{1}{2\pi}\int_{\mathbb{R}} P(\lambda)\, \mathrm{Re}\, \psi\big(\tfrac{1}{2} + \tfrac{i\lambda}{2}\big)\, \mathrm{d}\lambda \;=\; \tfrac{1}{8} + \tfrac{1}{4}\,\psi\big(\tfrac{1}{2}\big), \qquad P(\lambda) = \frac{\pi\lambda}{\sinh(\pi\lambda)},$$

by Fourier inversion followed by an elementary substitution. Two corrections to the earlier draft are recorded. The value 1/8 in that moment and the one-loop ladder moment $\mathcal{M}_1 = 1/8$ use different integration conventions (full line against half line), so their agreement is a proved coincidence of two separate evaluations, with no derivation linking them. And the vanishing of the Mellin bridge $(1 - 2^{-s})\Gamma(s)\zeta(s)$ at nontrivial zeros carries no information, since $\zeta(s)$ appears in it as a factor.

Keywords: celestial holography, conformal blocks, principal series, Legendre functions, Mehler–Fock transform.

Contents

1 Purpose and scope

A companion paper [1] derives the two-particle unitarity cut in celestial variables and shows that it is uniform on the round celestial sphere,

$$\mathrm{d}\Pi_2 = \frac{\mathrm{d}^2 z}{8\pi^2(1+|z|^2)^2}, \qquad z_6 = -\frac{1}{\bar{z}_5}, \tag{1}$$

with the two cut legs at antipodal points. To decompose that cut into $\mathrm{SL}(2,\mathbb{C})$ partial waves, one needs the scalar principal-series block as an explicit function, its behaviour under the shadow map, its size at large spectral parameter, and its moments against the weight $P(\lambda)$ that the same companion derives from phase space. This paper supplies those four things.

The earlier draft framed this material as the "kinematic factor" of a factorization in which $P(\lambda)$ served as a replacement for the Feynman loop measure. That framing is withdrawn in [2]. The mathematics below is untouched by the withdrawal, since none of it depends on the interpretation. What changes is the use to which it is put.

Conventions. $h = \frac{1}{2}(1+i\lambda)$ with $\lambda \in \mathbb{R}$ on the principal series; $\nu = h - 1 = -\frac{1}{2} + i\tau$, $\tau = \lambda/2$; $x(z) = 2/z - 1$; ψ is the digamma function. All numerical claims come from the archived script `verify_blocks_v2.py`.

2 Conical reduction, with its branch

Theorem 2.1 (Conical reduction). *For $h \notin \{0, -\frac{1}{2}, -1, \dots\}$ and $z \in (0,1)$,*

$$k_h(z) \;:=\; z^h\, {}_2F_1(h,h;2h;z) \;=\; c(h)\, Q_{h-1}\!\left(\frac{2}{z} - 1\right), \qquad c(h) = \frac{2\,\Gamma(2h)}{\Gamma(h)^2}, \tag{2}$$

where Q_ν is taken in the branch analytic on $(1,\infty)$ (the "type 3" or off-cut Legendre function), which is the branch selected by the argument $x = 2/z - 1 > 1$.

Proof. Quadratic transformation of ${}_2F_1$ into Legendre normal form, followed by identification of the recessive solution at $x \to \infty$. Verified at four points, including complex h on and off the principal series, to relative error below 3×10^{-41}. □

Remark 2.2 (The branch is load-bearing). Evaluated in the on-cut convention (Legendre Q with cut along $(-\infty, 1]$), (2) fails by 47% at $h = \frac{1}{2} + 0.7i$, $z = 0.3$, and by 98% at $h = 0.8$, $z = 0.25$. The earlier draft did not record the convention. Any reader reproducing the identity in a computer algebra system whose default is the on-cut branch will see it fail, so we state it here.

3 Shadow symmetry as Legendre degree symmetry

Theorem 3.1 (Shadow involution and the Mehler–Fock kernel). *Under $\Delta \mapsto 2 - \Delta$, equivalently $h \mapsto 1 - h$ and $\nu \mapsto -\nu - 1$, the reduction (2) maps Q_ν to $Q_{-\nu-1}$, and for $\nu = -\frac{1}{2} + i\tau$, $x > 1$,*

$$Q_{-\nu-1}(x) - Q_\nu(x) \;=\; i\pi \tanh(\pi\tau)\, P_\nu(x). \tag{3}$$

The shadow-odd part of the block is therefore the conical function $P_{-1/2+i\tau}$, the kernel of the Mehler–Fock transform.

Proof. Classical connection formula for Legendre functions of complex degree [3], specialized to the conical line. Verified at $(\tau, x) = (0.6, 3.0)$ and $(1.9, 5.5)$ to relative error below 1.3×10^{-41}. □

This is the reason the shadow transform behaves like an involution with a fixed line rather than a generic reflection: on the principal series it is a symmetry of the underlying Legendre equation, which depends on ν only through $\nu(\nu+1)$.

4 Exact prefactor and boundedness

Theorem 4.1 (Prefactor modulus). *For $h = \frac{1}{2}(1 + i\lambda)$, $\lambda > 0$,*

$$|c(\lambda)|^2 = \frac{4\,|\Gamma(1+i\lambda)|^2}{|\Gamma(\frac{1}{2}+\frac{i\lambda}{2})|^4} = \frac{2\lambda}{\pi}\coth\left(\frac{\pi\lambda}{2}\right). \tag{4}$$

Proof. Reflection gives $|\Gamma(1+i\lambda)|^2 = \pi\lambda/\sinh(\pi\lambda)$ and $|\Gamma(\frac{1}{2}+\frac{i\lambda}{2})|^2 = \pi/\cosh(\pi\lambda/2)$; then use $\sinh(\pi\lambda) = 2\sinh(\pi\lambda/2)\cosh(\pi\lambda/2)$. Verified to 10^{-31} at three values of λ. □

Theorem 4.2 (Boundedness on physical kinematics). *On the physical slice $z = e^{i\varphi}$, $\bar{z} = z^*$, with φ in a compact subset of $(0, 2\pi)$ avoiding $\varphi = 0$, the scalar block $g_{1+i\lambda}(z, \bar{z}) = k_h(z)k_h(\bar{z})$ obeys*

$$\big|g_{1+i\lambda}(z,\bar{z})\big| = |1-z|^{-1/2}\big(1 + O(\lambda^{-1})\big), \qquad \lambda \to +\infty. \tag{5}$$

No exponential growth survives: the rates $\pm\frac{1}{2}\mathrm{Im}\,\zeta$ of the two chiral factors are conjugate-odd and cancel.

Proof. Uniform large-degree asymptotics of Q_ν at fixed argument, applied to each chiral factor, combined with (4) in the form $|c|^2 = (2\lambda/\pi)(1 + O(e^{-\pi\lambda}))$. Checked numerically at $\varphi = 1.0$ and $\varphi = 2.2$: the ratio of $|g|$ to $|1-z|^{-1/2}$ equals 1.0059, 0.99989, 0.99914 at $\lambda = 4, 12, 30$, and 0.99836, 0.99892, 0.99899 respectively. □

Remark 4.3. The exponentially small correction $O(e^{-\pi\lambda})$ sits in the prefactor, not in the block. The earlier abstract described an "$e^{-\pi\lambda}$ envelope" for the block itself, which the numbers above contradict: at fixed z inside the disc, $|k_h(z)|$ approaches a constant near 0.72 as λ grows. Temperedness, in the sense that matters for spectral decomposition, is the boundedness statement (5).

5 The first moment, in closed form

Write $P(\lambda) = \pi\lambda/\sinh(\pi\lambda)$, which the companion paper derives as the phase-space factor $\Gamma(1+i\lambda)\Gamma(1-i\lambda)$ of the two-particle cut on the shadow locus [1]. Its Fourier pair is elementary:

$$P(\lambda) = \int_{\mathbb{R}} e^{i\lambda x}\,\frac{\mathrm{d}x}{4\cosh^2(x/2)}, \qquad \frac{1}{2\pi}\int_{\mathbb{R}} P(\lambda)\cos(\lambda y)\,\mathrm{d}\lambda = \frac{1}{4\cosh^2(y/2)}. \tag{6}$$

Theorem 5.1 (First moment against the digamma).

$$\frac{1}{2\pi}\int_{\mathbb{R}} P(\lambda)\,\mathrm{Re}\,\psi\big(\tfrac{1}{2}+\tfrac{i\lambda}{2}\big)\,\mathrm{d}\lambda \;=\; \frac{1}{8} + \frac{1}{4}\,\psi\big(\tfrac{1}{2}\big)\,. \tag{7}$$

Proof. Subtract the constant. By (6), $\frac{1}{2\pi}\int_{\mathbb{R}} P = \frac{1}{4}$, which produces the term $\frac{1}{4}\psi(\frac{1}{2})$. For the remainder use the Gauss integral representation $\psi(s) = -\gamma + \int_0^\infty \frac{e^{-t}-e^{-st}}{1-e^{-t}}\mathrm{d}t$, which gives

$$\operatorname{Re}\psi\big(\tfrac{1}{2}+\tfrac{i\lambda}{2}\big) - \psi\big(\tfrac{1}{2}\big) = \int_0^\infty \frac{e^{-t/2}\big[1-\cos(\lambda t/2)\big]}{1-e^{-t}}\,\mathrm{d}t. \tag{8}$$

Exchange the order of integration (the integrand is positive) and apply (6) at $y = t/2$, together with $e^{-t/2}/(1-e^{-t}) = 1/(2\sinh(t/2))$:

$$\frac{1}{2\pi}\int_{\mathbb{R}} P\left[\operatorname{Re}\psi - \psi(\tfrac{1}{2})\right] = \int_0^\infty \frac{1}{2\sinh(t/2)}\cdot\frac{1}{4}\Big[1-\operatorname{sech}^2(t/4)\Big]\mathrm{d}t. \tag{9}$$

Substituting $u = t/4$ and using $1/(\sinh u\cosh u) = \operatorname{sech}^2 u/\tanh u$, the integrand collapses to $\tanh u\,\operatorname{sech}^2 u$, so the remainder equals

$$\frac{1}{4}\int_0^\infty \tanh u\,\operatorname{sech}^2 u\,\mathrm{d}u = \frac{1}{4}\left[\frac{\tanh^2 u}{2}\right]_0^\infty = \frac{1}{8}. \tag{10}$$

Each step was checked numerically to at least 25 digits, and the final value matches the direct quadrature of (7) exactly at working precision. □

Remark 5.2 (The agreement of the two values of 1/8). The ladder moment of [2] is $\mathcal{M}_1 = \frac{1}{2\pi}\int_0^\infty P\,\mathrm{d}\lambda = \frac{1}{8}$, taken over the half line, while the $\frac{1}{8}$ in (7) comes from an integral over the full line. The earlier draft read the agreement as evidence that the one-loop measure appears inside the digamma moment. Section 6 shows that the two numbers really are the same quantity, and identifies what makes them so. The reason has nothing to do with loops.

6 Why the two values agree

Write $\hat{p}$ for the Fourier partner of P,

$$\hat{p}(x) = \frac{1}{2\pi}\int_{\mathbb{R}} P(\lambda)e^{-i\lambda x}\mathrm{d}\lambda = \frac{1}{4\cosh^2(x/2)}, \qquad \hat{p}(0) = \frac{1}{4}, \quad \hat{p}(\infty) = 0. \tag{11}$$

Both quantities in question are integrals of $\hat{p}$ against the kernel $1/\sinh$, so the question is what $\hat{p}$ does to that kernel.

Theorem 6.1 (Weight-shift relations and the resulting differential equation)**.** *Let* $P(\lambda) = \Gamma(1+i\lambda)\Gamma(1-i\lambda)$*. Then*

$$P(\lambda\mp i) = \frac{1\pm i\lambda}{\mp i\lambda}P(\lambda), \qquad -i(\lambda-i)P(\lambda-i) + i(\lambda+i)P(\lambda+i) = 4P(\lambda), \tag{12}$$

and $\hat{p}$ *satisfies, for all* x*, the first-order equation*

$$\tfrac{1}{2}\sinh(x)\,\hat{p}\,'(x) = \hat{p}(x) - \hat{p}(0). \tag{13}$$

Equation (13) *together with* $\hat{p}(0) = \frac{1}{4}$ *and* $\hat{p}(\infty) = 0$ *determines* $\hat{p}$ *uniquely.*

Proof. The first relation in (12) is the recursion $\Gamma(z+1) = z\Gamma(z)$ applied to each factor; the second follows by combining the two signs. Since $\lambda \mp i$ corresponds to $\Delta \mapsto \Delta \pm 1$ under $\Delta = 1 + i\lambda$, these are the integer weight shifts on the principal series. For (13), multiplication by $e^{\pm x}$ in the variable x is the shift $\lambda \mapsto \lambda \mp i$ on the transform side, and differentiation is multiplication by $-i\lambda$; so (12) transforms into $\frac{1}{2}\sinh(x)\hat{p}\,' = \hat{p} - \hat{p}(0)$, the constant arising because the shifted combination is integrable while $\hat{p}(0)$ alone is not. Direct check: with $\hat{p} = \frac{1}{4}\operatorname{sech}^2(x/2)$ one has $\hat{p}\,' = -\frac{1}{4}\operatorname{sech}^2(x/2)\tanh(x/2)$, so the left side is $\frac{1}{4}\tanh^2(x/2)$, which is $\hat{p}(0) - \hat{p}(x)$ up to sign. Verified at four values of x to absolute error below 1.3×10^{-32}. For uniqueness, separate variables: $\mathrm{d}\hat{p}/(\hat{p} - \frac{1}{4}) = 2\,\mathrm{d}x/\sinh x$ integrates to $\hat{p} - \frac{1}{4} = C\tanh^2(x/2)$, and $\hat{p}(\infty) = 0$ forces $C = -\frac{1}{4}$. □

Theorem 6.2 (The two numbers are one boundary value)**.**

$$\frac{1}{2\pi}\int_{\mathbb{R}} P(\lambda)\Big[\operatorname{Re}\psi(\tfrac{1}{2}+\tfrac{i\lambda}{2})-\psi(\tfrac{1}{2})\Big]\mathrm{d}\lambda = \int_0^\infty \frac{\hat{p}(0)-\hat{p}(x)}{\sinh x}\,\mathrm{d}x = -\frac{1}{2}\int_0^\infty \hat{p}\,'(x)\,\mathrm{d}x = \frac{\hat{p}(0)}{2} = \mathcal{M}_1. \tag{14}$$

Proof. The first equality is the Fourier step of Theorem 5.1, using $\operatorname{Re}\psi(\frac{1}{2}+\frac{i\lambda}{2}) - \psi(\frac{1}{2}) = \int_0^\infty (1-\cos\lambda x)/\sinh x\,\mathrm{d}x$. The second is (13) divided by $\sinh x$. The third is the fundamental theorem of calculus with $\hat{p}(\infty) = 0$. The last holds because $\mathcal{M}_1 = \frac{1}{2\pi}\int_0^\infty P\,\mathrm{d}\lambda = \frac{1}{2}\cdot\frac{1}{2\pi}\int_{\mathbb{R}} P\,\mathrm{d}\lambda = \frac{1}{2}\hat{p}(0)$, P being even. □

So the agreement is a theorem, and the mechanism is visible: the integrand of the digamma moment is an exact derivative, and what survives is the value of $\hat{p}$ at the origin. That same value, halved, is $\mathcal{M}_1$. Two integrals that look unrelated are two ways of evaluating $\hat{p}(0)/2$, and the equation that makes the first one collapse is the weight-shift relation (12) of the principal series. Nothing about loop amplitudes enters, which is the sense in which the earlier reading was wrong: the 1/8 is a property of the Gamma factors, and $\mathcal{M}_1$ inherits it.

Proposition 6.3 (The transform is a thermal window)**.** *$\hat{p}(x) = -n_F'(x)$, where $n_F(x) = (e^x+1)^{-1}$ is the Fermi–Dirac distribution at unit temperature in the variable x. Under the Bisognano–Wichmann reading of [2], x is boost rapidity and the temperature is $\beta = 2\pi$ in boost time.*

Remark 6.4 (Which archimedean factor each object comes from). $P = \Gamma(1+i\lambda)\Gamma(1-i\lambda)$ is built from Gamma factors of the type attached to a complex place, while $\operatorname{Re}\psi(\frac{1}{2}+\frac{i\lambda}{2})$ is the logarithmic derivative of $\Gamma(s/2)$ at $s = \Delta = 1 + i\lambda$, the type attached to a real place. Theorem 6.2 therefore pairs the two. We record the observation and stop there. No arithmetic consequence follows from it, and readers of the earlier drafts should note the vacuous zeta-zero check discussed in Sec. **??** as a cautionary case.

6.1 Higher moments

The mechanism above extends into a recursion. Put $f_k(y) := \frac{1}{2\pi}\int_{\mathbb{R}} P(\lambda)D(\lambda)^k e^{-i\lambda y}\mathrm{d}\lambda$ with $D(\lambda) = \operatorname{Re}\psi(\frac{1}{2}+\frac{i\lambda}{2}) - \psi(\frac{1}{2})$, so $f_0 = \hat{p}$ and $A_k := f_k(0) = \frac{1}{2\pi}\int_{\mathbb{R}} PD^k\mathrm{d}\lambda$. Define $T[f] := \int_0^\infty [f(0) - f(y)]/\sinh y\,\mathrm{d}y$. Then

$$A_{k+1} = T[f_k], \qquad T[f] = \tfrac{1}{2}f(0) - \int_0^\infty \frac{L[f](x)}{\sinh x}\mathrm{d}x, \qquad L[f] := \tfrac{1}{2}\sinh(x)f'(x) - f(x) + f(0), \tag{15}$$

and Theorem 6.1 says exactly that $L[\hat{p}]=0$, which is why the first step closes. Equivalently $A_k=(-1)^k(\mathcal{L}^k\hat{p})(0)$ for the operator $\mathcal{L}f(x)=\int_0^\infty[f(x+y)+f(x-y)-2f(x)]\,\mathrm{d}y/(2\sinh y)$, the generator of the symmetric Lévy process with jump density $1/(2\sinh|y|)$, whose characteristic exponent is D itself. For $k\geq 1$ the defect $L[f_k]$ does not vanish: in the transform variable it involves $\operatorname{Re}\psi(1+i\lambda)$, through the shifts

$$D(\lambda-i)-D(\lambda+i)=-\frac{2i}{\lambda},\qquad D(\lambda-i)+D(\lambda+i)=2\big[\operatorname{Re}\psi(\tfrac{i\lambda}{2})-\psi(\tfrac{1}{2})\big],\tag{16}$$

both verified to 15 digits. The moments themselves are striking, and we report them with their proof status marked.

k	A_k	Status	Check
0	1/4	proved	exact
1	1/8	proved (Thm. 6.2)	exact
2	1/8	**open**	40 digits
3	$\frac{1}{2}\log 2-\frac{3}{16}$	**open**	30 digits
4	$0.2315253305644009308\ldots$	no closed form found	n/a

Theorem 6.5 (The second moment)**.** $A_2=\frac{1}{8}=A_1$.

Proof. Substitute $\varphi=\tanh(x/2)$, under which $\mathrm{d}x/\sinh x=\mathrm{d}\varphi/\varphi$ and $a(x):=\hat{p}(0)-\hat{p}(x)=\frac{1}{4}\varphi^2$. Expanding the two cosine factors,

$$\frac{1}{2\pi}\int_{\mathbb{R}}P(\lambda)(1-\cos\lambda x)(1-\cos\lambda y)\mathrm{d}\lambda=a(x)+a(y)-\tfrac{1}{2}a(x+y)-\tfrac{1}{2}a(x-y),\tag{17}$$

so with φ,ψ the images of x,y and $s=\varphi\psi$, the addition rule $\tanh\frac{x\pm y}{2}=(\varphi\pm\psi)/(1\pm s)$ gives

$$A_2=\frac{1}{4}\int_0^1\!\!\int_0^1\left[\varphi^2+\psi^2-\frac{(\varphi+\psi)^2}{2(1+s)^2}-\frac{(\varphi-\psi)^2}{2(1-s)^2}\right]\frac{\mathrm{d}\varphi\,\mathrm{d}\psi}{\varphi\psi}.\tag{18}$$

The bracket collapses to $s^2[(\varphi^2+\psi^2)(s^2-3)+4]/(1-s^2)^2$. Setting $u=\varphi^2$, $v=\psi^2$ turns $\varphi\psi\,\mathrm{d}\varphi\,\mathrm{d}\psi$ into $\frac{1}{4}\mathrm{d}u\,\mathrm{d}v$, so

$$A_2=\frac{1}{16}\int_0^1\!\!\int_0^1\frac{(u+v)(uv-3)+4}{(1-uv)^2}\,\mathrm{d}u\,\mathrm{d}v.\tag{19}$$

Expand $(1-uv)^{-2}=\sum_{n\geq 0}(n+1)(uv)^n$ and integrate term by term, using $\int_0^1\int_0^1 u^av^b(uv)^n=[(a+n+1)(b+n+1)]^{-1}$. With $m=n+1$ the summand telescopes:

$$\sum_{m\geq 1}\left[\frac{2m}{(m+1)(m+2)}-\frac{6}{m+1}+\frac{4}{m}\right]=\sum_{m\geq 1}\left[\frac{4}{m}-\frac{8}{m+1}+\frac{4}{m+2}\right]=2,\tag{20}$$

the coefficients summing to zero so that the harmonic parts cancel and only the boundary terms $8-6$ survive. Hence $A_2=2/16=\frac{1}{8}$. Quadrature of (19) returns 2.0 at working precision. □

A second route reaches the same value and produces a tool needed below. Applying T to $\hat{p}$ in closed form gives the generator image

$$f_1(x)=\frac{1-c}{8}\left[2+\frac{(1+c)\log(1-c)}{c}\right],\qquad c=\tanh^2(x/2),\tag{21}$$

verified against direct quadrature of $-\mathcal{L}\hat{p}$ at four points to relative error below 3.1×10^{-30}, with $f_1(0) = \frac{1}{8} = A_1$. Then $A_2 = T[f_1] = \frac{1}{16}\int_0^1 [-1 + 2c - (1-c^2)c^{-1}\log(1-c)]\mathrm{d}c/c$, whose integrand has the cancellation-free form $\frac{5}{2} + c^{-2}[-\log(1-c) - c - \frac{c^2}{2}] + \log(1-c)$ and integrates to $\frac{5}{2} + \frac{1}{2} - 1 = 2$.

6.2 Status of the remaining moments

Normalizing $4P(\lambda)\mathrm{d}\lambda/2\pi$ to a probability measure, D has mean $A_1/A_0 = \frac{1}{2}$ and variance $A_2/A_0 - (A_1/A_0)^2 = \frac{1}{4}$ by Theorem 6.5, so its standard deviation equals its mean.

For A_3 the same machinery applies and the computation is longer. Two ingredients are in place. First, (21) reduces A_3 to a two-dimensional integral of elementary functions,

$$A_3 = \int_0^\infty \int_0^\infty \frac{b(x) + b(y) - \frac{1}{2}b(x+y) - \frac{1}{2}b(x-y)}{\sinh x \, \sinh y} \, \mathrm{d}x \, \mathrm{d}y, \qquad b := f_1(0) - f_1, \tag{22}$$

which quadrature confirms against $\frac{1}{2}\log 2 - \frac{3}{16}$ to eleven digits. Second, the logarithm in (21) splits under addition, since

$$1 - \tanh^2\!\Big(\frac{x+y}{2}\Big) = \frac{(1-\varphi^2)(1-\psi^2)}{(1+\varphi\psi)^2}, \tag{23}$$

so $\log(1-c)$ is additive up to $\log(1+\varphi\psi)$; the alternating denominator is the natural source of the $\log 2$. We have not carried the evaluation through, and record A_3 as open.

k	A_k	Status	Check
0	1/4	proved	exact
1	1/8	proved (Thm. 6.2)	exact
2	1/8	proved (Thm. 6.5)	exact
3	$\frac{1}{2}\log 2 - \frac{3}{16}$	open, reduced to (22)	30 digits
4	0.23152533056440093 08 . . .	no closed form found	n/a

References

[1] D. Toupin, "Loop amplitudes from unitarity cuts in celestial Mellin space," v19, 2026.

[2] D. Toupin, "The spectral weight $\pi\lambda/\sinh\pi\lambda$: derivation, modular form, and exact moments," v2, 2026.

[3] A. Erdélyi et al., *Higher Transcendental Functions*, Vol. 1, McGraw–Hill, 1953.

[4] H. T. Lam, S.-H. Shao, Phys. Rev. D98 (2018) 025020 [arXiv:1711.06138].

[5] "The celestial optical theorem," arXiv:2404.18898.